

Role of Smart Technologies to Reduced Human Effort

Wisdom Leaf Press
Pages number, 79–87
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<https://journals.icapsr.com/index.php/wlp>
DOI: 10.55938/wlp.v1i3.133



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Abstract

Artificial intelligence (AI) has garnered global interest due to the growth of big data and technology breakthroughs; as a result, major corporations such as Amazon, Google, Facebook, and Apple have made investments in AI-based services. Artificial Intelligence has the potential to enhance the quality of goods and services, reduce costs and increase efficiency. It can also benefit public entities such as financial markets and tax supervisory agencies. AI is predicted to completely change the financial industry and turn it into data-driven businesses. Artificial intelligence (AI) applications in the banking industry have produced encouraging outcomes, including higher revenue and profit through improved customer targeting and operational efficiency. However, big investment banks and credit card companies presently only use AI in a small number of backend functions, such as stock prediction and credit rating. The impoverished and financially disadvantaged can now obtain finance more easily thanks to digital financial strategies. Safe and dependable money management is critical in complicated domains like accounting and finance, where generative AI technologies like ChatGPT are being deployed. In the last five years, the use of AI has increased dramatically due to the proliferation of digital technology, algorithmic capabilities, and accessibility.

Keywords

Artificial Intelligence, Human Effort, Cybersecurity, Chatbots

Introduction

The rise of large data and recent technological advancements have sparked a global interest in artificial intelligence (AI). Some even promote “AI first” as a catchphrase and emblem for a significant upheaval of business structures that will create entirely new industries ^[1]. The fact that internet huge companies

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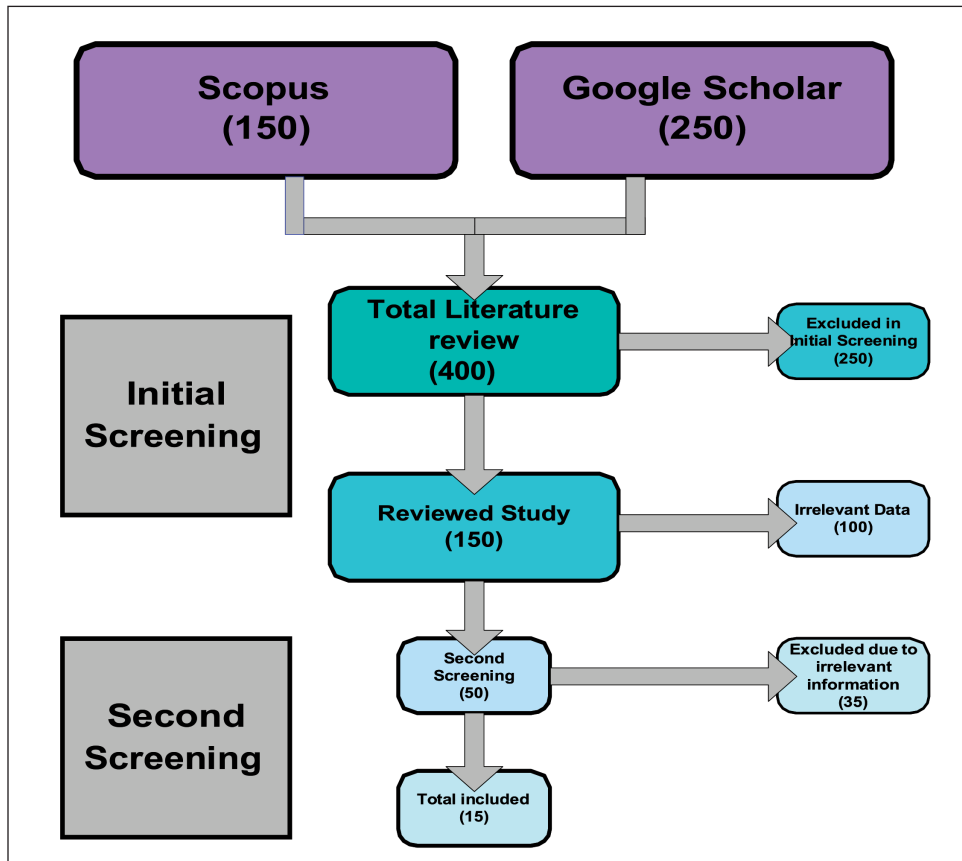


Fig 10.1: Literature sources

like Amazon, Google, Facebook and Apple are spending millions on Artificial intelligence to offer Artificial intelligence - based services and applications highlights the importance of Artificial intelligence for a wide range of various industries ^[2]. Using of Artificial intelligence can improve the quality of product and services for client due to a wide and larger analytical basis and information.

Secondly, AI could lead to higher efficiency and lower cost, example in the field of fraud detection and anti-money laundering measure. Moreover, AI technology might be helpful for the public entities in case of tax supervisory bodies or financial markets ^[8].

Third, AI may emerge as a major force behind innovation. While the exact nature of tomorrow's financial service provider is yet unknown, it is likely that AI will convert them into data- and AI-driven enterprises ^[9]. Using these AI application provides promising result for the banking sectors in the case of Banco Bilbao Vizcaya Argentaria (BBVA) in Spain, the use of ai leads an increase in the revenue and profit and a decrease in the cost through better customer targeting, the optimization of business operations, and the identification of better places for the branch networks, among other things ^[7]. However, ai in the finance and banking sector is currently applied in a limited number of backend servicers, like in stock prediction and credit rating by huge investment bank and credit card provider, which are highly

computerized [7]. Together, banks and non-bank organizations are utilizing digital financial techniques to increase financial accessibility for the underprivileged and financially disadvantaged. The fourth industrial revolution is bringing changes in the traditional banking sector built in the industrial revolution premised on paper and physical distribution of cash [3]. A study shows the function of ChatGPT and related generative AI technologies in the complex fields of accounting and finance, where safe and dependable money management is crucial. The importance of understanding these technologies and their effects in various disciplines is emphasized in this introduction. Financial industry transformation has been brought by generative AI, which can handle large datasets, automate tasks, and provide insightful data, as demonstrated by models such as ChatGPT [1]. The 2008 global financial crisis revealed several weaknesses in the financial systems and demonstrated the severe consequences that these failures had on governments, institutions, and society around the globe. Financial risk management is essential to preventing repeat occurrences of the same catastrophic events because of the vital connection that exists between the financial sector and the other economic sectors. Businesses must constantly adapt and manage financial risks in order to fulfill their vital role as society's financial intermediaries. Risk management in the financial sector is quickly using big data, computational frameworks, and AI algorithms in a variety of use cases as they become available. Financial risk management software integrated with ai is made with the goal of streamlining procedures to increase productivity, accuracy, and efficiency while cutting costs so that businesses can stay competitive [10]. AI adoption has accelerated significantly from the past five years as an outcome of the spread of digital technologies and major breakthroughs in algorithmic capabilities, giving access to richer data, and increasing computing power. Other than advancement and development in the areas of robotics, health care, aviation, education, medical diagnosis, transportation and businesses. These days most of the bank and financial institutions are using AI (Artificial intelligence) to change every aspect of the business by giving customize administrations, fulfil needs and make strides client encounter. Other than advancement and Artificial intelligence is most predominantly used in finance industry for Data Retrieval, Market Analytics, Insurance, Customer Service, Credit Scoring, Retail Lending, Process Automation, Risk Management and much more to strengthen the customer experience with improved technology in this digital era [6]. Budgeting and accounting are part of finance, and AI may help with strategic planning by predicting trends. AI's methodical examination of financial records helps auditing by highlighting areas of concern. taxes, intricate because of evolving legal requirements, advantages of AI computing, and guaranteeing compliance [5]. Many countries are exploring AI technologies and tools to promote financial inclusion with the goal of improving the well-being of households and communities. Through these contributions, using AI technologies is gaining ground as a way to advance financial inclusion. Advantages of using AI (Artificial intelligence) include making numerous financial services more convenient, extending the number of financial services that can be accessed digitally, producing data that can be used to identify ways in which customers could improve their welfare imposed from customers, building safer financial structures, reducing the costs of financial services such as remittances in order to make them more widely accessible, helping poor nations leverage their most abundant resources, which include a young population and a strong demand for financial services and boosting social welfare and economic growth by increasing the number of financially literate people relative to those who are not. Policymakers and financial service participants seeking to promote or adopt AI (Artificial intelligence) technologies should consider these benefits while being mindful about the associated risks [10].



Fig 10.2: Advantages of AI in

Objectives

- To examine the role of artificial intelligence (AI) in reducing human efforts in the finance sector.
- To investigate several uses of AI in finance, including as risk management, fraud detection, algorithmic trading, customer support, and regulatory compliance.
- To emphasize the advantages of using AI in finance, including more productivity, better decision-making, lower costs, and better customer service.
- To determine potential future paths and areas for AI growth in the finance sector.
- To offer case studies and illustrations showing the effective application of AI-driven financial solutions.

Literature Review

AI is being rapidly adopted in the financial sector, leveraging big data to enhance predictive power and decision-making processes. It has the potential to reduce time and cost of service, support client-centric strategies, and increase protection against fraudulent activities. Furthermore, the utilization of AI also brings new challenges and risks for the financial system, which require consideration of regulatory aspects. This paper tackles the possible benefits and potential challenges of AI in finance. The paper discusses the challenges, opportunities and regulatory developments of AI in the financial industry. It identifies possible benefits of AI in customer experience, risk management, cost reduction, and access to finance, and it also considers the risks and concerns it brings, while the implementation of artificial

intelligence optimization algorithms in the financial sector, exposing their advantages and necessity. There are many ways in which AI can benefit finance, and one of the primary ones is the time and cost it can save on various aspect of financial services — from customer support to portfolio management. AI works ideally with the finance sector, as it's capable of analyzing data and identifying patterns much faster than the human equivalent, in addition to automating the initial groundwork that is the very bane of human existence. This means that AI can deal with the big data side of financial operations, clear trails for human workers to look at the more creative and intentional part of the field. AI is also useful for finance in that it can help to personalize and make more client-centric company strategies. By allowing AI to keep an eye on customer behavior and individual needs, a company can build up a stock of data that can be used to create diverse payout and insurance quotes; allow financial institutions to monitor client preferences and decisions for signs of suspicious activity and create quicker systems of credit and repayment that are molded to individual needs personalized to individual clients. All of these things will help a client feel trusted and secure, encouraging them to stay with one company rather than flitting from one to the other as soon as they see a better deal. It is hard to ignore the potential benefits of AI in the financial industry. AI systems can improve customer experience, as with the automated chatbots and virtual assistants that provide instant customer service. They can make customer service faster, smarter and more personal, leading to improved overall satisfaction and retention rates. AI algorithms can also help financial institutions more accurately assess risks and develop effective risk management strategies. Traditional methods simply aren't capable of analyzing and making use of such a vast amount of data in real time. AI technology can analyze and interpret thousands — even millions — of data points in real time, facilitating the proactive identification and resolution of potential issues. These opportunities and challenges, regulators have a key role in shaping the responsible deployment of AI in the financial sector. Regulatory frameworks must evolve with the technological landscape to ensure consumer protection, data privacy and market integrity. Moreover, regulators must collaborate with industry stakeholders to develop standards and guidance to facilitate the responsible and ethical deployment of AI, ensuring transparency, fairness and accountability in the algorithmic decision-making process.

Research Methodology

The nature of this study is conceptual as it attempts to aggregate data on AI in the financial sector and do a literature review but also very original in terms of studying related issues. There are still three main areas that need examining in terms of practice though: efforts reduced by AI in human labour; AI's rich application in the field of finance and potential future for AI in finance -- all without any clear conclusions so far!

Conduct a comprehensive review of existing literature on AI in the finance sector, focusing on academic journals, industry reports, and reputable online sources like google scholar. Find and select keywords and phrases relevant to AI in finance, “artificial intelligence,” “financial sector,” “automation,” “AI in finance sector ” and “future trends”. Review any studies published within the last 10 years in order to ensure that your information is relevant and current.

Data Collection and Analysis

Gather relevant research articles, papers, and reports identified during the literature review process. Organize the collected data into three major themes: (a) Role of AI in reducing Human Effort, (b) Diverse applications of AI in the financial sector, and (c) Potential Future of AI in Finance. From all the selected

research articles, extract the key findings, insights, methodologies and recommendations that they provided to inform this research paper. Summarize and analyses the collected data to identify common trends, patterns, challenges, and opportunities across the three thematic areas.

Role of AI To Reducing Human Effort

From automating routine tasks and improving decision-making to enhancing customer service and personalizing experiences, optimizing resource allocation, and even enabling innovation, AI reduces

Table 10.1: Important Literature Review

Technology	Year	Finding
AI	2020	In conclusion, this study demonstrates the significant impact of AI (artificial intelligence) on digital financial inclusion. By leveraging AI technologies, fintech companies can address key barriers to financial access and empower low-income earners, women, youth, and small businesses to participate in the mainstream financial market. AI facilitates risk detection and management, reduces information asymmetry, enhances customer support through chatbots, and strengthens fraud detection and cybersecurity measures. It is therefore recommended that financial institutions, non-financial institutions, and governments worldwide adopt and expand the use of AI tools and applications to facilitate greater financial inclusion with minimal challenges and maximum benefits for vulnerable groups ^[3] .
AI	2023	The paper finds that generative AI, such as ChatGPT, is transforming finance and accounting by streamlining processes, improving decision-making, and reducing operational costs. However, challenges related to ethics, data privacy, regulatory compliance, and accountability remain significant hurdles. Despite these challenges, the transformative potential of AI in these sectors is evident, necessitating ongoing research and development efforts to address these issues and fully realize its benefits ^[5]
AI	2019	The study finds that while AI has been a research field for years, recent advancements and data availability have made it commercially applicable. Major tech companies heavily invest in AI, but its implementation in finance remains limited. Through expert interviews, the study identifies drivers and inhibitors of successful AI applications, highlighting the importance of role models, process competencies, and addressing ethical concerns ^[1]
AI	2019	The study shows that AI has become commercially viable due to recent tech advancements and abundant data. Major companies like Google and Amazon invest significantly in AI, highlighting its global relevance. In finance, AI is promising but implementation is limited. Through expert interviews, the study identifies key factors for successful AI adoption, including role models, process competencies, and addressing ethical concerns ^[1] .
AI	2020	The study reveals that AI is underutilized in customer interactions in commercial banking. AI can mitigate lending losses, boost payment security, automate compliance, and refine customer targeting. Challenges include integration, user acceptance, privacy, and documentation, prompting a suggested research agenda for behavioral finance ^[7] .

human effort. Machine learning algorithms and natural language processing enable AI to automate numerous repetitive and tedious tasks, from data entry to administrative tasks, freeing up valuable time that can be dedicated to more strategic pursuits. By quickly and efficiently analyzing vast amounts of data, AI helps provide critical insights that can speed the decision-making process across a wide range of industries, markets and sectors using AI-driven analytics tools. Round-the-clock AI-powered chatbots and virtual assistants can rapidly respond to customer inquiries as well as deliver personalized suggestions, and AI-driven personalization can optimize user engagement across a variety of platforms. Other innovative uses of AI include predictive maintenance to allow manufacturing businesses to anticipate machinery malfunctions or failures and resolve malfunctions before they occur, and route optimization in the transportation business that lowers the time, cost and resources required to move products from first sale to final destination. In short, AI amplifies human advantage by automating processes and improving efficiency.

Diverse Applications of AI in Financial Sector

Artificial intelligence (AI's) impact crosses various avenues in the financial sector. Data analysis has significantly improved risk management, fraud detection is done with the help of machine learning algorithms, algorithmic trading is enhanced by optimizing trading strategies, chatbots allow for an interactive and more personalized AI-driven customer service, automation streamlines regulatory compliance, underwriting has been revolutionized by credit scoring models that use AI and machine learning to more accurately assess creditworthiness, tools for tracking trends in real time provide cutting edge market analysis, AI optimizes asset allocation for portfolio management, and more powerful, cutting-edge AI—an emerging trend—keeps institutions ahead of rapidly evolving markets by providing real-time insights. AI's applications showcase its potential in finance: transformative, revolutionizing and improving processes so that they are more efficient and less costly with better-informed decision-making and improved compliance and customer satisfaction (Forbes).

Advantages of AI in Finance

The marriage of Artificial Intelligence (AI) and finance delivers numerous benefits, including operational efficiency gains from the automation of tasks such as data entry and fraud detection. Resource allocation is greatly optimized, and human capital is freed up for more strategic tasks. Real-time analysis of huge datasets also allows financial institutions to gain invaluable insights into customer behavior and market trends, enabling them to make more informed decisions and better their investment strategies. AI also facilitates predictive analytics, allowing banks and other financial institutions to anticipate market movements and better manage any risks that are involved, resulting in a more resilient financial operation. Artificial intelligence (AI) not only reduces operational costs but ensures that federally mandated requirements are adequately addressed. Regulatory strain and penalties associated with the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010, is the perfect example. In a highly dynamic stage of finance, AI artificial intelligence (AI) fulfils the necessity to interpret the regulation and its many amendments by creating highly configurable, rules-based engines that can prompt changes in regulations and help financial services keep their competitive edge. Furthermore, AI (Artificial intelligence) driven customer service solutions, such as chatbots and virtual assistants, transformative transaction experiences and build customer satisfaction levels as brand extensions by offering personalized recommendations

and transform transaction experiences. Essentially, all this to say: artificial intelligence (AI) not only reduces operational costs but meets regulatory compliance requirements and enhances competitive advantage within the highly dynamic stage of finance.

Finance Potential Future of AI in Finance

The future looks bright for AI in finance. From revolutionizing risk assessment to reducing back-office costs, we are already seeing the impact of AI on various aspects of finance. AI- powered systems can digest vast amounts of data in real time and provides more accurate predictions and better decision making. As machine-learning and natural language processing, among other technologies, continue to develop, so too will the ways that AI can contribute to the finance sector. The impact goes beyond the profession, as well. Advanced AI systems can also power a new generation of robotics-advisors, offering more personalized financial advice based on an individual's personal finances and goals – effectively democratizing access to financial services. But the future of AI in finance isn't simply a question of when and what. How – and for that matter, whether – the systems will comply with ethical standards, regulations, and the need for transparency all matter just as much.

Recommendation

- AI reduces the burden for big financial institutions and banks as it can perform typical calculations in seconds.
- AI reduces the risk of financial failure like in the past example in 2008 as it can predict future by the data given to it.
- Use of AI in the financial sector reduce the operational costs and improve decision- making of financial institution.
- AI taking over the jobs of people which may be engaged in that field for their living hood.

Conclusion

In conclusion, the adoption of Artificial Intelligence (AI) in the financial services industry has dramatically decreased the amount of human labor and has disrupted traditional means. AI- based systems automate various tasks including data entry, transactions processing and compliance checks, allowing employees to redirect their focus on strategic and value-added activities. Tremendous amounts of financial data are reviewed by machine learning algorithms at a speed and accuracy unmatched, enabling predictive analytics for risk management, fraud detection or investment strategies. Similarly, AI-driven chatbots and virtual assistants instantly provide support and recommend various solutions, greatly improving overall user satisfaction and retention for the industry. AI-based algorithms optimize trading decisions, portfolio management and algorithmic trading, increasing returns and minimizing risk.

Furthermore, the adoption of AI technologies in the finance sector offers an array of benefits. Organizations will be empowered with better operational efficiency, more effective decision- making processes and the ability to drive innovation like never before. The industry also needs to navigate a number of challenges, however. These include data privacy concerns, bias in algorithms, and the need for the ongoing monitoring and regulation of the technologies to ensure their responsible and ethical

implementation. Despite these, the transformative potential of AI in finance is hard to overestimate, ushering in a new era of efficiency, accuracy and customer-centricity.

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